

TIRUCHIRAPPALLI, India

WMO: 433440

Lat: 10.765N

Lon: 78.710E

Elev: 288

StdP: 14.54

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	68.1	69.6	58.6	74.3	87.6	60.6	79.8	85.8	19.1	82.0	18.4	82.0	4.2	20	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	18.9	102.6	78.8	101.8	78.6	100.3	78.3	81.9	95.0	81.0	93.6	80.5	92.8	8.5	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	152.8	85.9	78.5	149.6	85.4	77.4	144.2	84.6	46.1	96.0	45.1	94.1	44.4	93.3	93.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.6	20.4	18.3	DB	64.9	105.2	1.6	1.8	63.7	106.5	62.8	107.5	61.9	108.5	60.7	109.8	(4)
(5)			WB	62.0	84.5	1.7	2.6	60.8	86.4	59.8	87.9	58.8	89.3	57.5	91.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	85.3	78.7	81.7	86.0	90.0	90.4	89.7	89.0	87.8	86.8	84.0	80.6	78.5	(6)	
	DBStd	4.96	1.92	2.40	2.87	2.93	3.16	2.37	2.14	2.31	2.49	2.97	2.47	2.10	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	12874	888	887	1118	1199	1251	1191	1209	1172	1104	1055	917	883	(10)	
	CDD65	7399	423	467	653	749	786	741	744	707	654	590	467	418	(11)	
	CDH74	91337	3465	4819	8315	10723	11334	10459	10292	9451	8412	6649	4220	3198	(12)	
	CDH80	48139	1201	2221	4554	6508	6987	6189	5899	5143	4359	2849	1327	902	(13)	
(14) Wind	WSAvg	8.5	7.8	7.3	6.2	5.9	8.4	11.7	13.5	11.1	8.5	6.2	6.9	7.9	(14)	
(15) Precipitation	PrecAvg	36.0	0.7	0.4	0.4	1.4	2.3	1.4	2.4	3.4	6.7	7.2	5.4	4.1	(15)	
	PrecMax	51.7	6.8	4.9	2.5	6.3	6.3	4.8	8.0	8.7	16.3	13.7	13.0	19.0	(16)	
	PrecMin	19.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.4	1.9	1.8	0.2	0.3	(17)	
	PrecStd	8.1	1.6	1.0	0.6	1.6	1.5	1.4	2.2	2.4	3.2	3.4	3.6	4.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.7	96.4	102.0	104.4	105.6	103.6	102.1	100.6	100.4	98.3	91.8	89.8	(19)	
		MCWB	73.8	75.0	76.0	79.1	79.7	79.1	77.3	77.3	77.4	77.6	76.8	74.8	(20)	
	2%	DB	89.0	93.5	99.8	102.5	103.2	100.8	100.2	98.8	98.5	95.3	89.9	87.8	(21)	
		MCWB	73.2	74.0	76.2	78.7	79.4	78.5	77.4	77.4	77.1	77.9	76.4	74.7	(22)	
	5%	DB	87.6	91.7	97.8	100.8	101.8	99.3	98.5	97.1	96.7	93.3	88.2	86.2	(23)	
		MCWB	72.9	73.6	75.9	78.5	79.4	78.3	77.3	77.2	77.0	77.7	76.0	74.1	(24)	
	10%	DB	86.0	90.0	95.4	98.9	99.9	97.7	96.7	95.3	94.8	91.2	86.4	84.6	(25)	
		MCWB	72.5	73.3	75.5	78.4	79.3	78.0	77.3	77.1	77.1	77.5	75.9	73.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.6	78.6	81.0	83.1	83.3	82.7	81.7	80.9	80.9	80.2	79.9	78.9	(27)	
		MCDB	85.2	88.5	92.0	96.5	96.7	97.5	95.3	92.2	90.7	89.0	86.3	84.3	(28)	
	2%	WB	76.3	77.2	79.4	81.6	81.9	81.0	80.1	79.7	79.9	79.7	78.9	77.6	(29)	
		MCDB	84.0	87.2	90.8	94.2	95.5	95.2	93.5	91.9	90.2	88.3	85.3	83.2	(30)	
	5%	WB	75.3	76.4	78.7	80.8	81.0	79.9	79.1	79.0	79.2	79.1	78.2	76.6	(31)	
		MCDB	82.6	86.1	90.1	92.6	94.4	94.0	92.5	91.4	89.6	87.5	84.3	82.3	(32)	
	10%	WB	74.5	75.6	77.9	80.0	80.4	79.0	78.2	78.2	78.6	78.6	77.6	75.8	(33)	
		MCDB	81.8	85.0	88.9	91.7	93.6	93.0	91.7	90.5	89.0	86.8	83.4	81.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	16.2	19.0	20.6	19.2	18.9	17.3	16.5	16.5	16.6	14.2	12.1	13.4	(35)	
		MCDBR	17.6	20.3	22.2	21.2	21.0	19.2	18.5	18.3	18.8	17.3	14.9	15.3	(36)	
	5% WB	MCWBR	5.3	5.5	5.3	5.1	5.8	5.6	5.3	5.4	5.7	5.4	4.8	5.4	(37)	
		MCWBR	14.4	17.3	19.5	19.3	19.3	18.5	17.1	17.0	16.7	14.8	12.0	12.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.458	0.457	0.503	0.524	0.572	0.488	0.505	0.494	0.482	0.529	0.507	0.486	(40)	
		taud	2.192	2.190	2.067	2.057	1.935	2.206	2.147	2.186	2.222	2.068	2.112	2.137	(41)	
	Ebn at Noon		269	274	263	254	238	255	252	259	264	250	252	257	(42)	
		Edh at Noon	46	48	55	55	61	46	49	48	46	53	49	48	(43)	
(44) All-Sky Solar Radiation	RadAvg		1590	1861	1997	1979	1901	1784	1730	1764	1774	1531	1319	1339	(44)	
(45)	RadStd		75	80	110	77	108	98	87	85	93	94	131	107	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.61	-0.29	N/A	+1.82	N/A	-0.54	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.65	N/A	N/A	+1.62	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page